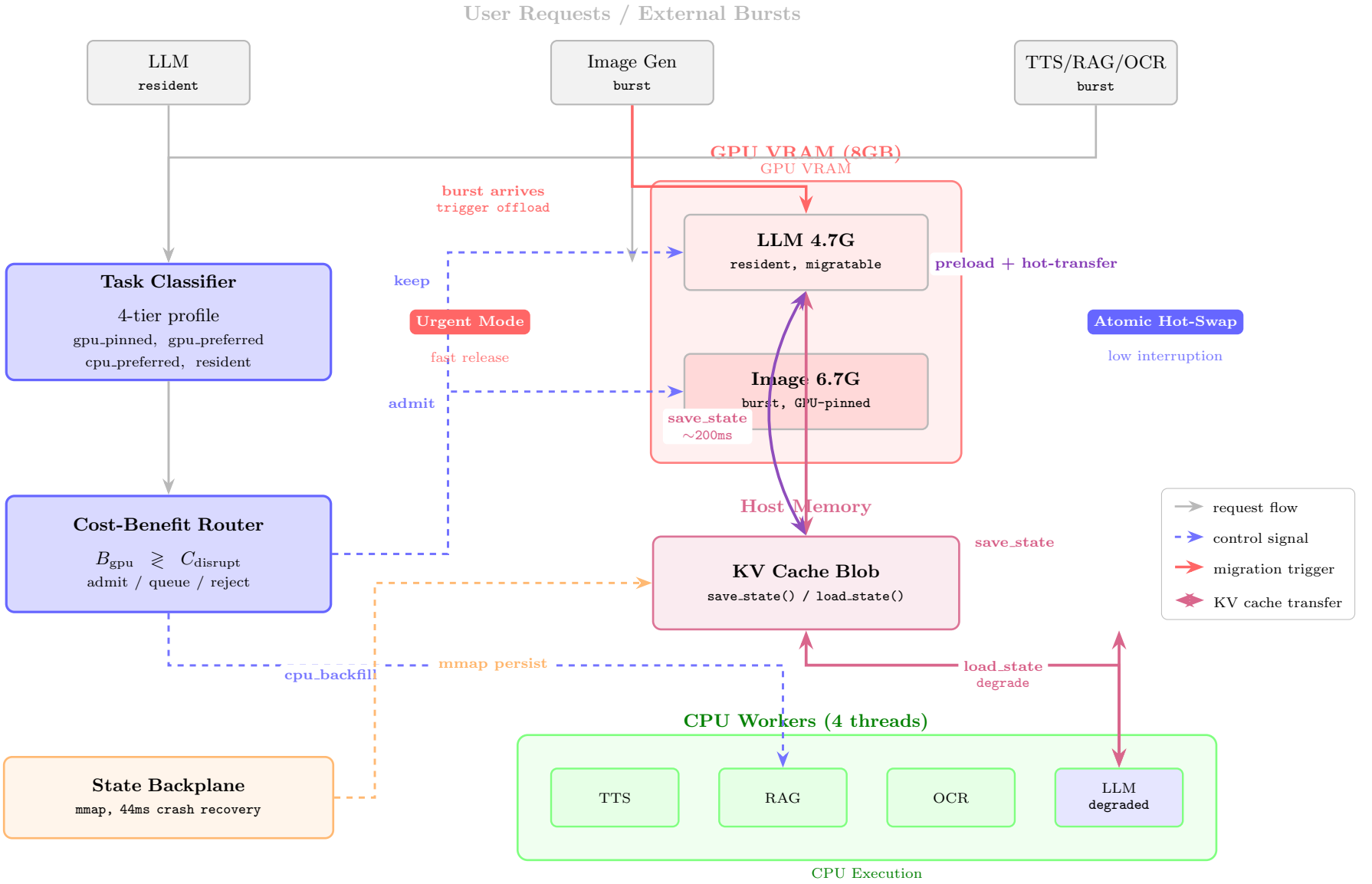

User Requests / External Bursts
LLM
resident
Image Gen
burst
TTS/RAG/OCR
burst
GPU VRAM (8GB)
GPU VRAM
burst arrives
trigger offload
LLM 4.7G
resident, migratable
preload + hot-transfer
Task Classifier
4-tier profile
gpu_pinned, gpu_preferred
cpu_preferred, resident
keep
Urgent Mode
fast release
admit
Atomic Hot-Swap
low interruption
Image 6.7G
burst, GPU-pinned
save_state
~200ms
Host Memory
Cost-Benefit Router
$B_{gpu} \gtrless C_{disrupt}$
admit / queue / reject
save_state
KV Cache Blob
save_state() / load_state()
request flow
control signal
migration trigger
KV cache transfer
cpu_backfill
mmap persist
load_state
degrade
CPU Workers (4 threads)
State Backplane
mmap, 44ms crash recovery
TTS
RAG
OCR
LLM
degraded
CPU Execution